

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science that builds systems able to perform tasks that normally need human intelligence. These tasks include understanding language, recognizing images, making decisions, and learning from experience.

The term was first used in 1956 at the Dartmouth Conference, which is widely considered the birth of AI as a research field. Since then, AI has gone through several waves of excitement and slowdown, often called AI summers and AI winters.

Today, AI is used in search engines, voice assistants, recommendation systems, medical imaging, fraud detection, and self-driving research. Most of the recent progress comes from machine learning and deep learning.

Machine Learning Basics

Machine learning is a branch of AI where computers learn patterns from data instead of following hand-written rules. There are three main types: supervised learning, unsupervised learning, and reinforcement learning.

In supervised learning, the model learns from labeled examples. For instance, a house price model is trained on houses with known prices. Common algorithms include linear regression, decision trees, and random forests.

In unsupervised learning, the model finds structure in unlabeled data. Clustering customers into groups based on their buying habits is a typical example. In reinforcement learning, an agent learns by trial and error using rewards and penalties.

A good model must generalize well to new data. Overfitting happens when a model memorizes the training data and performs badly on unseen data. Techniques such as cross-validation and regularization help prevent it.

Large Language Models and RAG

Large Language Models (LLMs) are neural networks trained on huge amounts of text. They can write, summarize, translate, and answer questions. However, they can sometimes produce wrong answers confidently, which is called hallucination.

Retrieval-Augmented Generation (RAG) reduces hallucination by giving the model relevant documents at question time. The system first searches a knowledge base for the most relevant passages, then passes them to the LLM together with the question.

A typical RAG pipeline has four steps: split documents into chunks, convert each chunk into an embedding vector, store the vectors in a vector database, and retrieve the closest chunks for each user question.

Embeddings are numerical representations of text. Two pieces of text with similar meaning have embeddings that are close together, which allows semantic search instead of simple keyword matching. Popular vector databases include ChromaDB, Pinecone, and FAISS.

Evaluating and Deploying AI Projects

Evaluation depends on the task. Classification models are measured with accuracy, precision, recall, and F1 score. Regression models use mean absolute error or root mean squared error. RAG systems are judged on retrieval quality and answer faithfulness to the source documents.

Deployment turns a notebook into a real product. A common approach is to wrap the model in an API using FastAPI, package it with Docker, and host it on a cloud platform. Simple demos can be deployed quickly using Streamlit Community Cloud.

After deployment, monitoring is important. Teams track response time, error rates, cost per request, and user feedback. API keys must be stored in environment variables and never committed to GitHub.